

Course Outcomes:**Knowledge
Level, KL**

Upon the completion of the course, the student will be able to:

CO 1	Apply concepts of Digital Binary System and implementation of Gates.	K ₃
CO2	Analyze and design of Combinational logic circuits.	K ₄
CO3	Analyze and design of Sequential logic circuits with their applications.	K ₄
CO4	Implement the Design procedure of Synchronous & Asynchronous Sequential Circuits.	K ₃
CO5	Apply the concept of Digital Logic Families with circuit implementation.	K ₃

Detailed Syllabus**UNIT I**

Digital System And Binary Numbers: Number System and its arithmetic Signed binary numbers, Logic simplification and combinational logic design: Binary codes, code conversion, review of Boolean algebra and Demorgans theorem, SOP & POS forms, Canonical forms, Karnaugh maps method up to five variable, Don't care conditions, POS simplification, NAND and NOR implementation, Quine McClusky method (Tabular method).

UNIT II

Combinational Logic: MSI devices like Magnitude comparator, Multiplexers, Demultiplexers, Decoders, Encoders. Multiplexed display, half and full adders, subtractors, serial and parallel adders, BCD adder

UNIT III

Sequential Logic And Its Applications: Storage elements: latches & flip flops, Characteristic Equations of Flip Flops, Flip Flop Conversion, Shift Registers, Ripple Counters, Synchronous Counters, Other Counters: Johnson & Ring Counter.

UNIT IV

Synchronous & Asynchronous Sequential Circuits: Analysis of clocked sequential circuits with state machine designing, State reduction and assignments, Design procedure. Analysis procedure of Asynchronous sequential circuits, circuit with latches, Design procedure, Reduction of state and flow table, Race-free state assignment, Hazards.

UNIT V

Memory & Programmable Logic Devices: Digital Logic Families: DTL, DCTL, TTL, ECL & CMOS etc., Fan Out, Fan in, Noise Margin; RAM, ROM, PLA, PAL; Circuits of Logic Families, Interfacing of Digital Logic Families, Circuit Implementation using ROM, PLA and PAL

Text Books:

1. M. Morris Mano and M. D. Ciletti, "Digital Design", Pearson Education.
2. Digital Circuits and Design, S. Salivahanan, Oxford University Press
3. David J. Comer, "Digital Logic & State Machine Design", Oxford University Press.
4. RP Jain, "Modern Digital Electronics", McGraw Hill Publication.
5. A. Anand Kumar, "Fundamental of Digital Circuits," PHI 4th edition, 2018.
6. D.V. Hall, "Digital Circuits and Systems," McGraw Hill, 1989.